Oluwaseyifunmi Olowe

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Education

University of Southern California — Los Angeles, CA

Bachelor of Science in Computer Engineering and Computer Science — ***May 2029***

- **Relevant Coursework:** Embedded Systems, Distributed Systems/IoT, Data Structures and Algorithms, Object-Oriented Programming, Analog Circuits

Technical Skills

Languages: C, C++, Python, Assembly
Systems & Tools: Embedded Linux, FreeRTOS, Git, PyTorch, GDB, CMake, CUDA, Docker, AWS, KiCad
Protocols & Peripherals: CAN, SPI, I2C, UART/USART, ADC, PWM, GPIO, TCP/IP, HTTP, MQTT
Lab Equipment: Oscilloscope, Power Supply, Signal Generator, Digital Multimeter, Spectrum Analyzer

Experience

Embedded Software Engineer — Los Angeles, CA

USC Formula SAE Electric — *Sep 2025 – Present*

- Develop **STM32F4** accumulator-management firmware in **C/FreeRTOS** using **CAN, SPI, ADC-DMA**, flash, and watchdogs for real-time battery safety monitoring.
- Build a **Python test-automation** GUI to decode DBC signals, replay CAN logs, and detect invalid data.
- Implement a BQ79600 **SPI driver** with CRC validation, timeouts, response parsing, and daisy-chain bring-up.
- Write **7 host-side C test suites** with **100+ assertions** validating CAN protocols, battery logic, and ADC data.

Electrical Engineering Intern — Los Angeles, CA

California DREAMS — *May 2026 – Aug 2026*

- Designed a **4-layer audio amplifier PCB** in KiCad within a 2 in × 2 in footprint, from schematic to fabrication.
- Designed and tested stereo-summing, **low-pass filtering**, gain, and **power-decoupling** circuits to amplify bass.
- Verified frequency responses, gain, and supply voltages using an **oscilloscope, multimeter**, and signal generator.
- Reworked a heart-rate monitor PCB using **SMD soldering** and identified **4 design errors** via schematic review.

Undergraduate Researcher — Los Angeles, CA

University of Southern California — *May 2026 – Jul 2026*

- Optimized hardware-aware **spiking neural networks (SNNs)** for deployment on superconducting SFQ chips.
- Built **Python/PyTorch simulations** to model SNN accuracy, operation count, and SFQ hardware constraints.
- Pruned and quantized the network to fit a **3.4 × 3.9 mm** SFQ chip, **reducing fan-in and circuit complexity.**
- Cut inference cost by **5×** (1.27M to 254K operations) with a **3.65% accuracy gain**, reducing on-chip energy use.

Projects

GPU ThermaGuard | *C++, ESP-IDF, FreeRTOS, CUDA, NVML, AWS IoT*

- Engineered a hardware-in-the-loop **GPU thermal controller** linking an ESP32-S3 to a **cloud NVIDIA T4 GPU** through **AWS IoT Core**.
- Built a **C++ NVML agent** publishing GPU temperature, power, utilization, and clock telemetry over MQTT.
- Wrote a **CUDA burn kernel** with temperature-based duty cycling, cutting GPU power from **70 W to 33 W**.
- Developed **FreeRTOS control tasks** exchanging telemetry and commands over **mutual TLS MQTT**.

Hawkeye | *Embedded C, FreeRTOS, Git, Sensor Fusion, Motion Control*

- Built an embedded **positioning system** in **C/FreeRTOS** to find light sources and control motion.
- Fused accelerometer and gyroscope data with a **complementary filter** for real-time orientation estimation.
- Combined orientation and phototransistor data to estimate target bearing, confidence, and **motion commands**.
- Wrote a register-level **I2C driver** and FreeRTOS tasks for concurrent **sensing, perception, and actuation**.

Leadership & Involvement

USC Viterbi Impact — Jan 2026 – Present

- Mentor a cohort of **6 students**, teaching robotics and Python programming to support early STEM education.

National Society of Black Engineers — Aug 2025 – Present

- Active member of NSBE, participating in professional development workshops and community initiatives.